INTRODUCTION TO ORGANIC CHEMISTRY

NCHE31205

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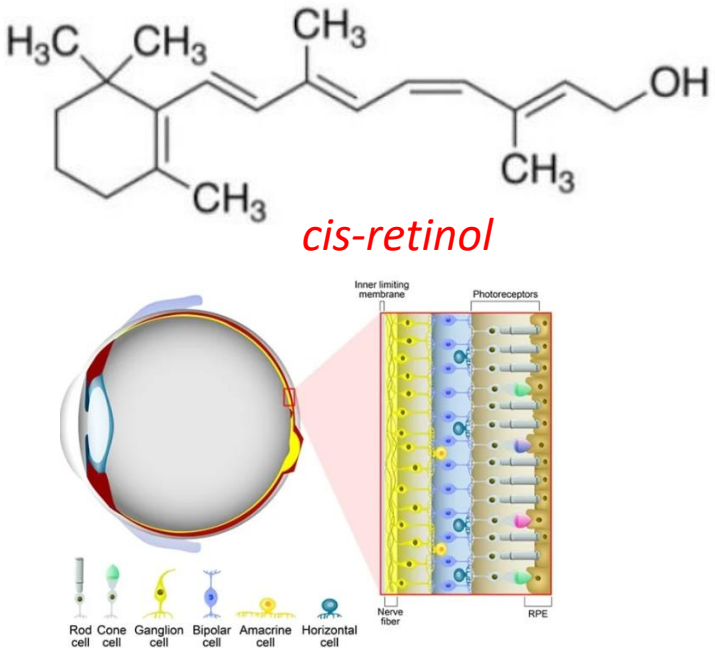
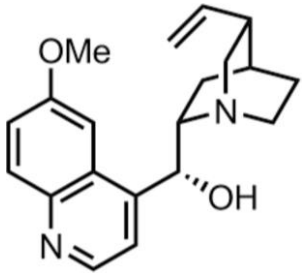
SEMESTER II

2024

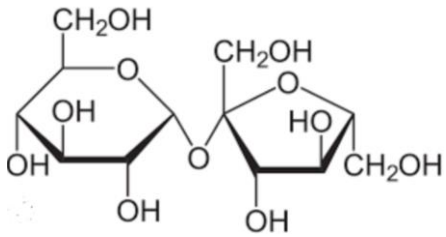
- The study of compounds containing carbon is called *organic chemistry.*
- Compounds that contain carbon and hydrogen, often in combination with nitrogen, oxygen, and/or other elements are known as *organic compounds*.
- Most biologically relevant compounds are organic.



cis-retinol

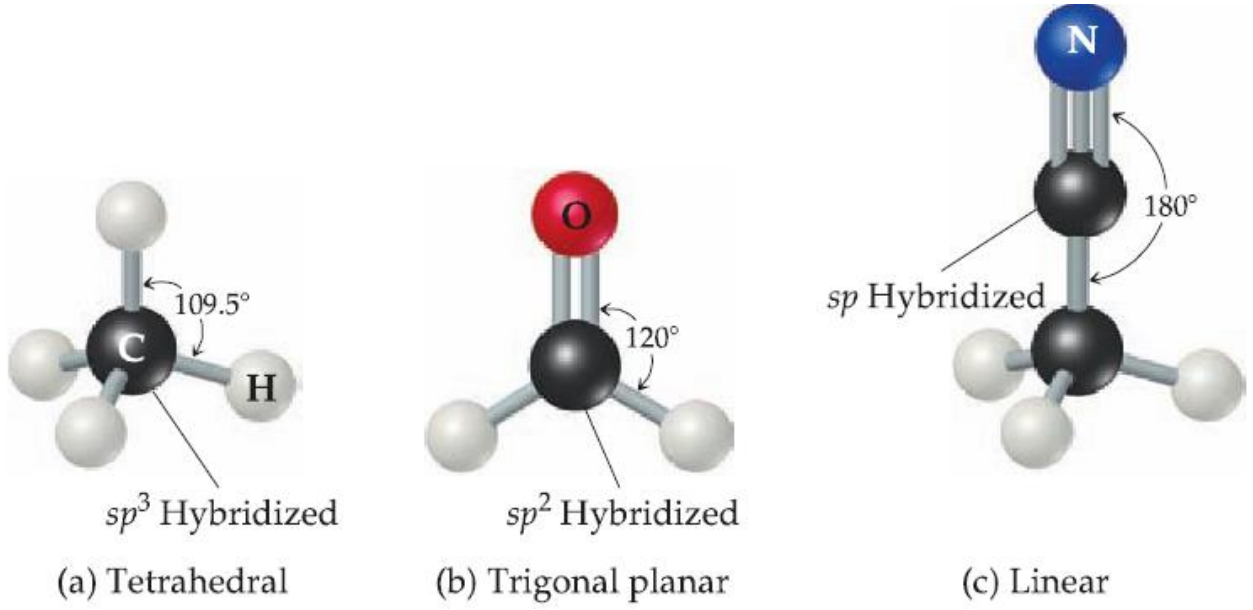


Quinine



sucrose

- Carbon has four valence electrons; [He]$2s^2 2p^2$.
- It therefore forms four bonds in virtually all of its compounds.
- When all four bonds are single bonds, the molecular geometry is tetrahedral – sp^3 hybridized.
- One double bond – trigonal planar – sp^2 hybridized.
- One triple bond – linear – sp hybridized.



Drawing molecules

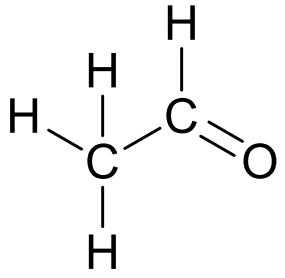
Structural formulae: Shows how the atoms in the organic molecule are bonded together.

- Different types of structural formulae:

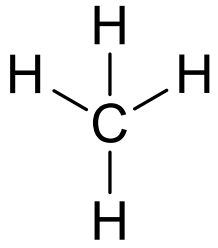
1) Displayed formulae/ Lewis structures:

- This shows all the bonds between atoms in the molecule as individual lines.
- All atoms (including carbon and hydrogen) in the organic molecule are shown.

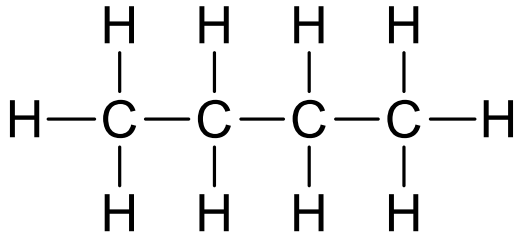
- For example, the displayed formula for the following compounds:



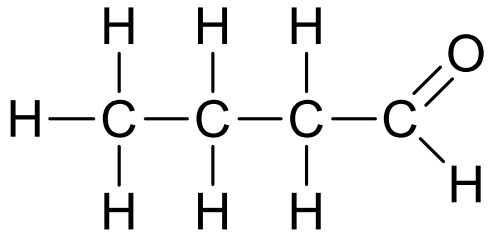
ethanal



methane



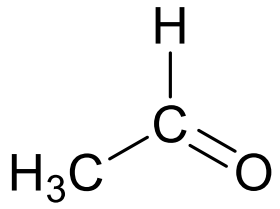
butane



butanoic acid

2) **Condensed structural formulae:**

- Condense the C—H bonds in order to simplify the displayed formula. For example by writing CH_3 or CH_2 instead of showing these bonds.
- For example:



ethanal

CH_4 methane

$CH_3—CH_2—CH_2—CH_3$ butane

$CH_3—CH_2—CH_2—C(=O)OH$ Butanoic acid

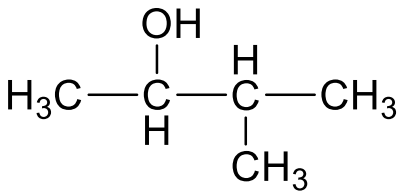
- The structural formula can even be condensed further to minimise space by avoiding displaying of any bonds.
- For example;

CH_3COH

$CH_3CH_2CH_2CH_3$

$CH_3CH_2CH_2COOH$

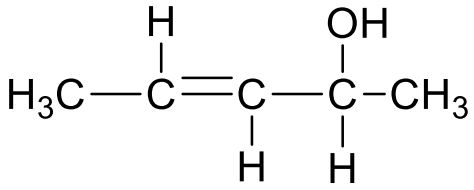
- However, does not show presence of multiple bonds, or arrangement of atoms in the molecule.
- Substituents are indicated using brackets;
- For example, the condensed formula for 2-hydroxy-3-methylbutane:



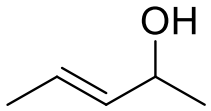
is: $CH_3CH(OH)CH(CH_3)CH_3$

3) **Skeletal formulae:**

- Simplify the representation of organic compounds by omitting C's and H's in the formula.
- Shows only the skeleton of the compound by representing C—C bonds as lines.
- H's are omitted.
- Any heteroatom (oxygen, nitrogen, halogen) is shown.
- Any functional group is shown, attached to the skeleton.
- For example, the skeletal formula for:

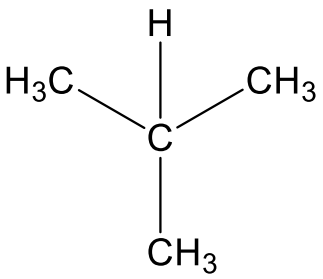


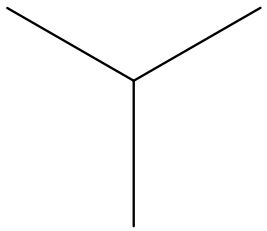
is:

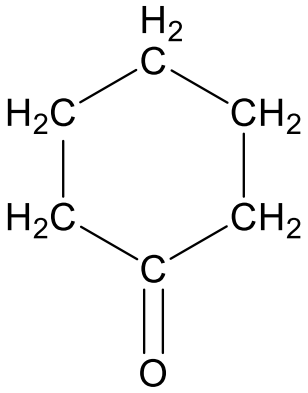


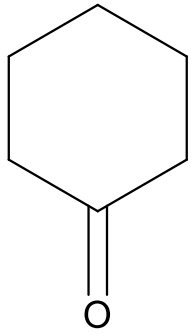
Condensed displayed formula

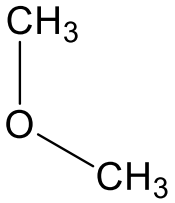
Skeletal formula

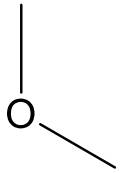












Hydrocarbons

- The simplest class of organic molecules is the *hydrocarbons*, which consist only of carbon and hydrogen.
- Contain C—C and C—H bonds.
- C—C bonds form the skeleton / backbone of the molecule, while the hydrogen atoms are on the surface or ‘skin’ of the molecule.
- The key feature of hydrocarbons is the presence of stable C—C bonds.
- These bonds are non-polar, and have a high bond strength.

- Hydrocarbons can be grouped into four categories based on the types of carbon-carbon bonds in their molecules:

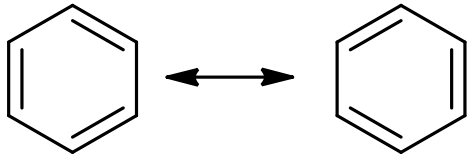
1. Alkanes
2. Alkenes
3. Alkynes
4. Aromatic hydrocarbons.

Saturated and Unsaturated Hydrocarbons

- Saturated hydrocarbons are those than contain only C—C single bonds.
- They are *saturated* with hydrogen.
- For example, alkanes and cycloalkanes.
- Unsaturated hydrocarbons are those that contain at least one C=C double bond or C≡C triple bond.
- For example, alkenes, alkynes, cycloalkenes.

Aliphatic and Aromatic Hydrocarbons

- Aromatic hydrocarbons are those that contain a benzene ring in their structure.
- Benzene has *delocalised* (spread out) pi electrons – this gives rise to the two resonance structures of benzene.



Resonance

- They usually have very pleasant smells, hence the name.
- Most organic compounds are aromatic.
- An aliphatic compound is one that consists of no rings; only a straight / branched chain.
- A cyclic compound is one that contains a non-aromatic cyclic ring in its structure.

Naming Organic compounds

- Chemists should be able to name organic compounds, so that their structures can be drawn.
- Chemists have adopted a set of rules for naming compounds established by an organisation called the International Union of Pure and Applied Chemistry (IUPAC).
- The naming of an unbranched chain consists primarily of 2 parts;

1) The prefix: this indicates the number of carbon atoms in the chain
2) The suffix (ending): this indicates the functional group/ nature of the compound.

Prefixes

- The prefix is determined by the number of carbon atoms in the unbranched chain.

Carbon Atom (#)	Root Name
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

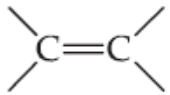
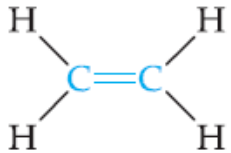
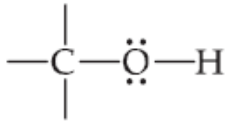
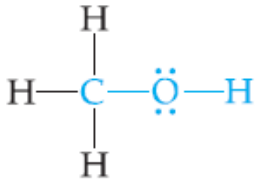
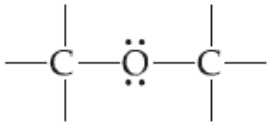
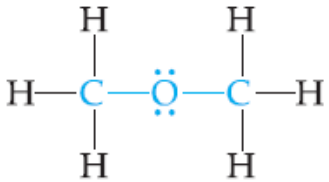
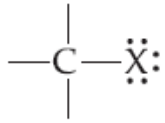
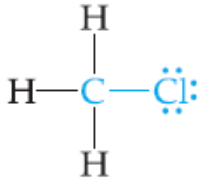
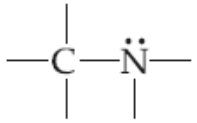
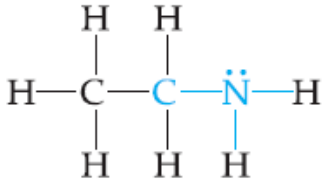
Memorize these!!

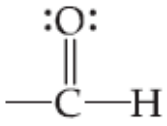
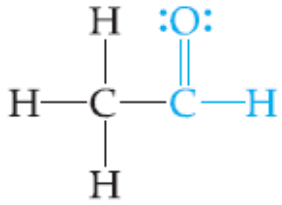
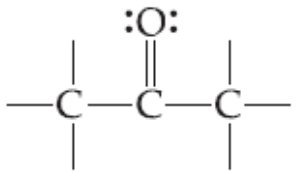
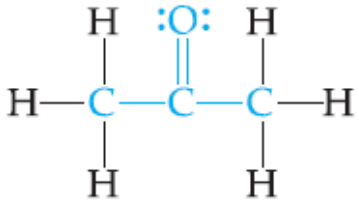
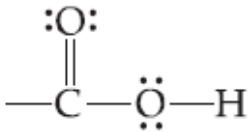
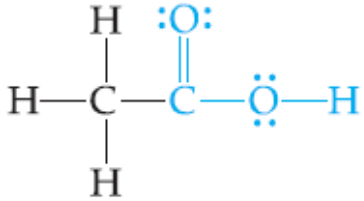
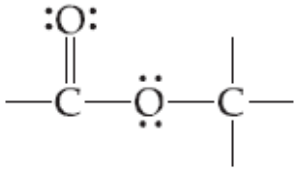
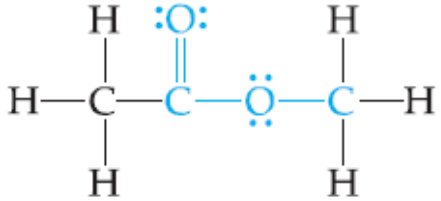
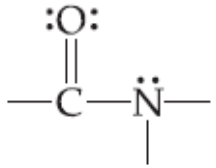
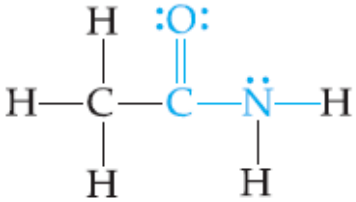
Suffixes

- The suffix is determined by the *functional group* of the molecule.

- The reactivity of organic compounds can be attributed to certain atoms or groups of atoms within the molecule. These groups of atoms make up a functional group.

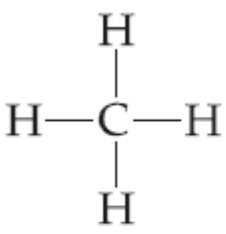
- A site of reactivity within an organic molecule is known as a *functional group* – it controls how the molecule behaves or functions.

Functional Group	Type of Compound	Suffix	Example	Systematic Name
	Alkene	*-ene*		Ethene
	Alkyne	*-yne*		Ethyne
	Alcohol	*-ol*		Methanol
	Ether	*ether*		Dimethyl ether
 (X = halogen)	Haloalkane	*halo-*		Chloromethane
	Amine	*-amine*		Ethylamine

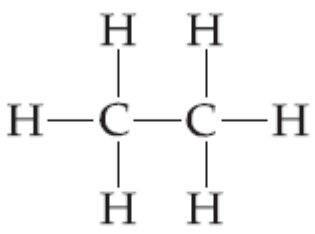
	Aldehyde	*-al*		Ethanal
	Ketone	*-one*		Propanone
	Carboxylic acid	*-oic acid*		Ethanoic acid
	Ester	*-oate*		Methyl ethanoate
	Amide	*-amide*		Ethanamide

Alkanes

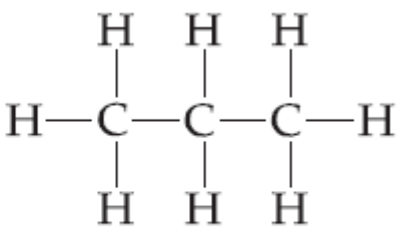
- These are hydrocarbons that contain only single bonds. Thus each carbon atom forms four single bonds.
- Names of alkanes end in '*ane*'.
- To name the alkane, add '*ane*' to the prefix corresponding to number of carbon atoms;



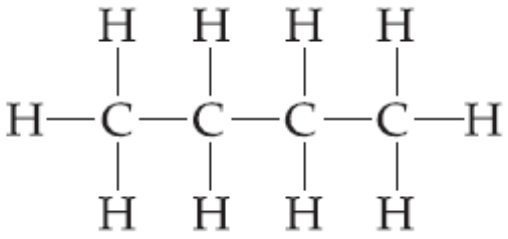
Methane



Ethane



Propane



Butane

Lewis Structures/Displayed formulae

- General formula for alkanes ; C_nH_{2n+2}

- The formulas for alkanes are often written in the form of *condensed structural formulas.*

- For example, the molecular formula for propane is C_3H_8. The condensed structural formula is $CH_3CH_2CH_3$.

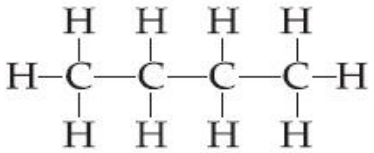
First few members of the alkanes and their corresponding condensed structural formulae

Molecular Formula	Condensed Structural Formula	Name	Boiling Point (°C)
CH_4	CH_4	Methane	−161
C_2H_6	CH_3CH_3	Ethane	−89
C_3H_8	$CH_3CH_2CH_3$	Propane	−44
C_4H_{10}	$CH_3CH_2CH_2CH_3$	Butane	−0.5
C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$	Pentane	36
C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$	Hexane	68
C_7H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	Heptane	98
C_8H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	Octane	125
C_9H_{20}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	Nonane	151
$C_{10}H_{22}$	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	Decane	174

These are all straight chain hydrocarbons or straight chain alkanes

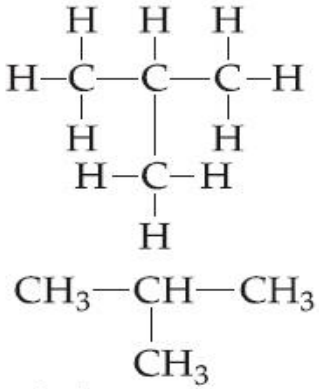
Structural isomers of alkanes

- Alkanes consisting of four or more carbon atoms can also form *branched chains.*



$CH_3CH_2CH_2CH_3$

Butane
mp −138 °C
bp −0.5 °C



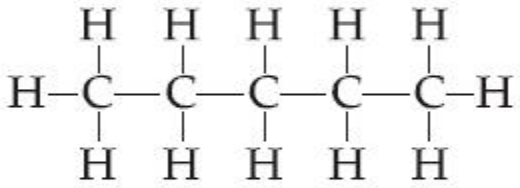
$CH_3—CH(CH_3)—CH_3$

Isobutane
(2-methylpropane)
mp −159 °C
bp −12 °C

Compounds with the same molecular formula but different structural formula (different arrangement of atoms) are known as *structural isomers.*

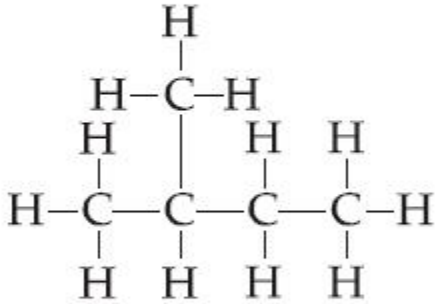
For example, there are two structural isomers for butane.

Pentane has three structural isomers



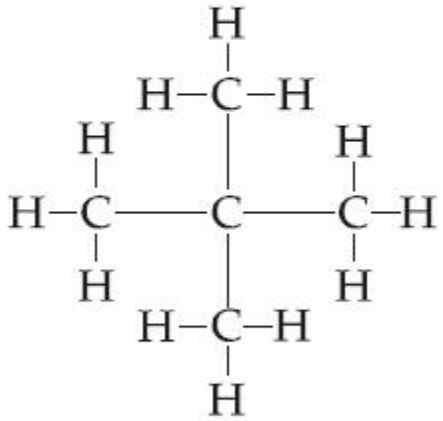
$CH_3CH_2CH_2CH_2CH_3$

Pentane
mp −130 °C
bp +36 °C



$CH_3-CH(CH_3)-CH_2-CH_3$

Isopentane
(2-methylbutane)
mp −160 °C
bp +28 °C



$CH_3-C(CH_3)_2-CH_3$

Neopentane
(2,2-dimethylpropane)
mp −16 °C
bp +9 °C

- Practice Exercise 1:

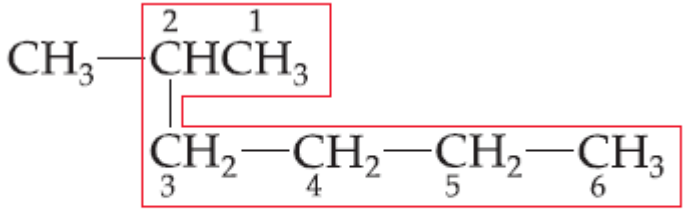
a) Draw the displayed formula for pentane.

b) Draw the displayed formulae for any 4 structural isomers of pentane.

Naming Structural Isomers

- The following steps summarize the procedure for naming structural isomers:

1. *Find the longest continuous chain of carbon atoms, and use the name of this chain as the base name of the compound*. the longest chain may not always be written as a straight line;



2-Methyl*hexane*

Groups attached to the main chain are called *substituents*.

2. *Number the carbon atoms in the longest chain, beginning with the end of the chain nearest to the substituent*. The chain is numbered from the end that gives the lowest number for the substituent position.

3. Name and give the location of each substituent group. A substituent group that is formed by removing an H from an alkane is called an alkyl group. Named by replacing the '*ane*' ending of the alkane with '*yl*'.

Names and structural formulae of alkyl substituents

Name	Condensed Structural Formula	Name	Condensed Structural Formula
methyl	$-CH_3$	isobutyl	$-CH_2CH(CH_3)$
ethyl	$-CH_2CH_3$	*sec*-butyl (abbreviation for "secondary")	$-CH(CH_3)CH_2CH_3$
propyl	$-CH_2CH_2CH_3$	*tert*-butyl (abbreviation for "tertiary")	$-C(CH_3)_3$
isopropyl	$-CH(CH_3)_2$		
butyl	$-CH_2CH_2CH_2CH_3$		

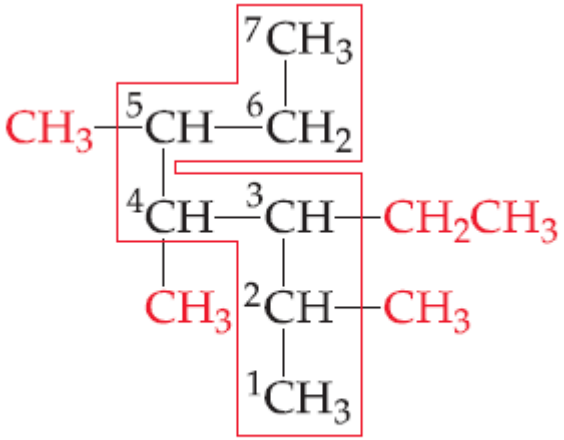
Secondary – carbon bonded to two other carbons

Tertiary – carbon bonded to three other carbons

Note: if the substituent is an OH group, it is given the name 'hydroxy'.

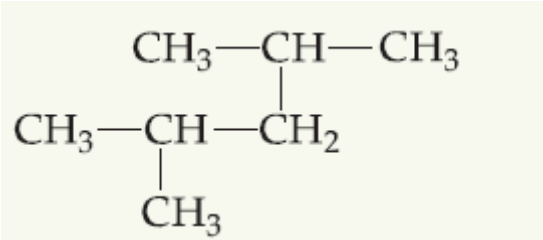
If the substituent is a halogen, it is given the name 'halo'. For example a chlorine substituent is known as 'chloro'.

4. When two or more substituents are present, list them in alphabetical order. When there are two or more of the same substituent, use a prefix (*di-, tri-, tetra-*) in front of the substituent name.

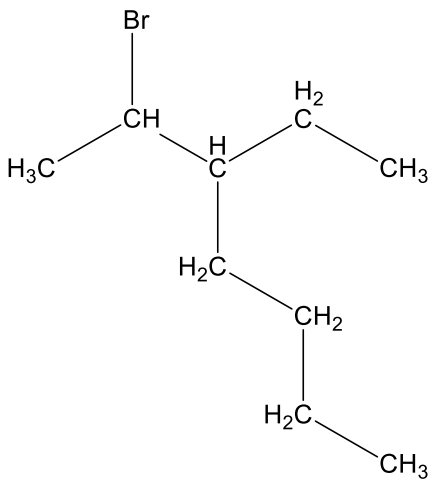


3-Ethyl-2,4,5-trimethylheptane

- Practice exercise 2: a) name the following alkane:



b) Name the following organic compound:



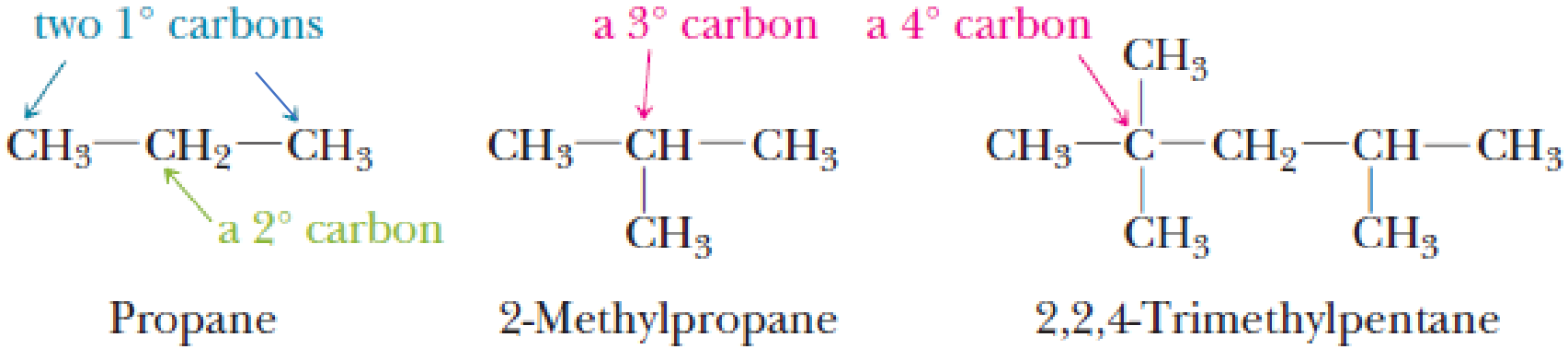
c) Draw the condensed displayed formula for 3-ethyl-2-methylpentane.

d) Draw the skeletal formula for the above alkane.

e) Name the structural isomers for hexane that you drew in Practice exercise 1.

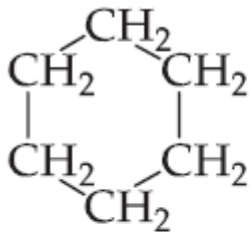
Classification of carbons

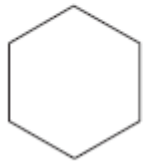
- We classify carbons as primary (1°), secondary (2°), tertiary (3°), or quaternary (4°) based on the number of carbon atoms bonded to that particular carbon.



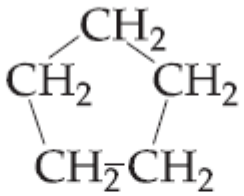
Cycloalkanes

- Cycloalkanes are saturated hydrocarbons that contain carbon atoms joined in a ring.
- General formula: C_nH_{2n}

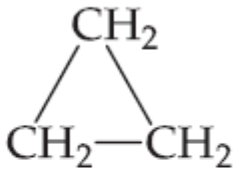




Cyclohexane

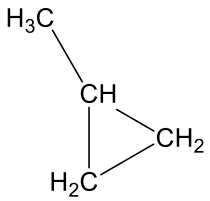


Cyclopentane



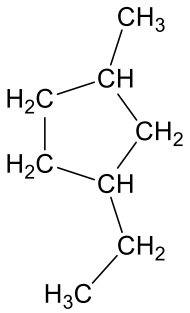
Cyclopropane

- To name cycloalkanes, prefix the name of the corresponding open-chain hydrocarbon with *cyclo-*, and name each substituent on the ring.
- If there is only one substituent, there is no need to give it a number.



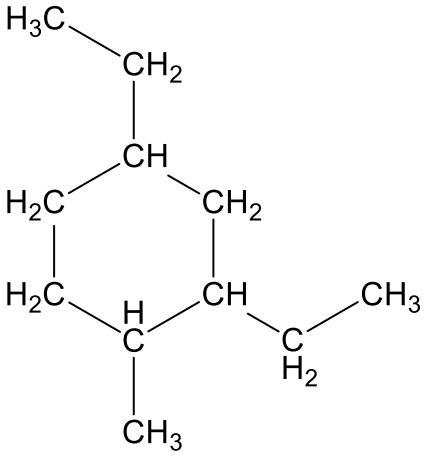
Methylcyclopropane

- If there are two substituents, number the ring by beginning with the substituent of lower alphabetical order. Number in such a way that you get the *lowest set of numbers*.

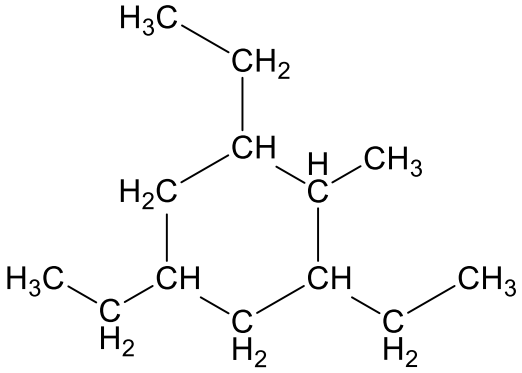


1-ethyl-3-methylcyclopentane
NOT 1-ethyl-4-methylcyclopentane!!

- If there are three or more substituents, number the ring so as to give them the lowest set of numbers, and then list the substituents in alphabetical order.



1,3-diethyl-4-methylcyclohexane



1,3,5-triethyl-2-methylcyclohexane

Sources of alkanes

- The three major sources of alkanes are: natural gas, coal, and petroleum oil (the three fossil fuels).
- Natural gas consists of approximately 90 -95% methane, 5-10% ethane, and a mixture of other relatively low-boiling alkanes.
- Through the process of thermal cracking, alkanes separated from these sources can be converted into unsaturated hydrocarbons (more useful).

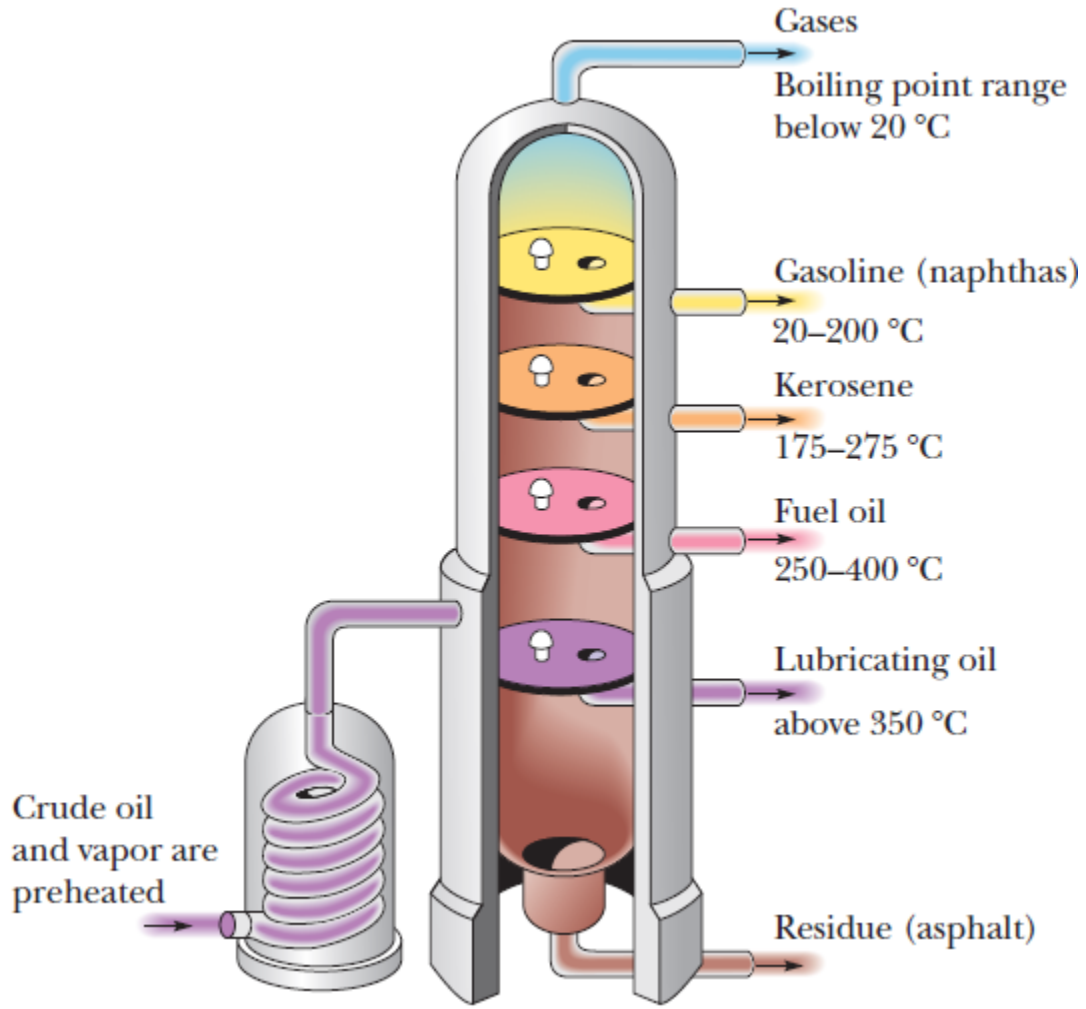
$$\underset{\text{Ethane}}{CH_3CH_3} \xrightarrow[\text{(thermal cracking)}]{800\text{–}900\,^\circ C} \underset{\text{Ethylene}}{CH_2{=}CH_2} + H_2$$

- Catalytic cracking of alkanes takes place at lower temperatures, in the presence of catalysts.

$$\underset{\text{Hexane}}{CH_3CH_2CH_2CH_2CH_2CH_3} \xrightarrow[-H_2]{\text{catalyst}} \underset{\text{Cyclohexane}}{\text{⬡}} \xrightarrow[-3H_2]{\text{catalyst}} \underset{\text{Benzene}}{\text{⬡}}$$

- Petroleum oil is a thick viscous liquid containing a mixture of thousands of compounds, most of which are hydrocarbons.
- Petroleum, and petroleum-derived products fuel automobiles, trains, and aircrafts.
- They also provide greases and lubricants for machinery.
- In addition, petroleum and natural gas contribute 90% of the raw materials which are used in industry to make dyes, pharmaceuticals, perfumes, plastics, synthetic fibres, etc.

- But first, the petroleum must be refined.
- This complex mixture of hydrocarbons must be separated into fractions. This is achieved by fractional distillation.



Heated to temperatures as high as 370 to 425 °C.
Each fraction contains a mixture of hydrocarbons that boils at a specific range.

1. Gases boiling below 20 °C are taken off at the top of the distillation column. This fraction is a mixture of low-molecular-weight hydrocarbons, predominantly propane, butane, and 2-methylpropane, substances that can be liquefied under pressure at room temperature. The liquefied mixture, known as liquefied petroleum gas (LPG), can be stored and shipped in metal tanks and is a convenient source of gaseous fuel for home heating and cooking.
2. Naphthas, bp 20 to 200 °C, are a mixture of C_5 to C_{12} alkanes and cycloalkanes. Naphthas also contain small amounts of benzene, toluene, xylene, and other aromatic hydrocarbons The light naphtha fraction, bp 20 to 150 °C, is the source of straight-run gasoline and averages approximately 25% of crude petroleum. In a sense, naphthas are the most valuable distillation fractions, because they are useful not only as fuel, but also as sources of raw materials for the organic chemical industry.
3. Kerosene, bp 175 to 275 °C, is a mixture of C_9 to C_{15} hydrocarbons.
4. Fuel oil, bp 250 to 400 °C, is a mixture of C_{15} to C_{18} hydrocarbons. Diesel fuel is obtained from this fraction.
5. Lubricating oil and heavy fuel oil distill from the column at temperatures above 350 °C.
6. Asphalt is the black, tarry residue remaining after the removal of the other volatile fractions.

- Coal is mostly carbon (C).
- To make use of coal as a raw material for organic synthesis, steam is passed over the coal.
- This results in the formation of synthetic gas – a mixture of carbon monoxide and hydrogen.

$$\underset{\text{Coal}}{C} + H_2O \xrightarrow{\text{heat}} CO + H_2$$

- Synthetic gas can then be used to manufacture methanol and acetic acid:

$$CO + 2H_2 \xrightarrow{\text{catalyst}} \underset{\text{Methanol}}{CH_3OH}$$

$$\underset{\text{Methanol}}{CH_3OH} + CO \xrightarrow{\text{catalyst}} \underset{\text{Acetic acid}}{CH_3\overset{\overset{O}{\|}}{C}OH}$$

Properties of Alkanes

- The most important property of alkanes and cycloalkanes is their almost complete lack of polarity – limits their reactivity.
- Their boiling points are generally lower than those of other compounds of similar molecular weight – weaker intermolecular forces.
- A branched alkane generally has a lower boiling point than the corresponding unbranched alkane of the same molecular weight.
- As branching increases, boiling point decreases.
- This is because as branching increases, the molecule becomes more compact, reducing surface area for intermolecular forces of attraction.

- Alkanes containing 1 to 4 carbons are gases at room temperature.
- Those containing 5 to 17 carbons are colorless liquids.
- Those containing 18 carbons and above are waxy, white solids - for example, the petroleum jelly used in cosmetics which is a mixture of high-molecular weight alkanes.

Reaction mechanisms

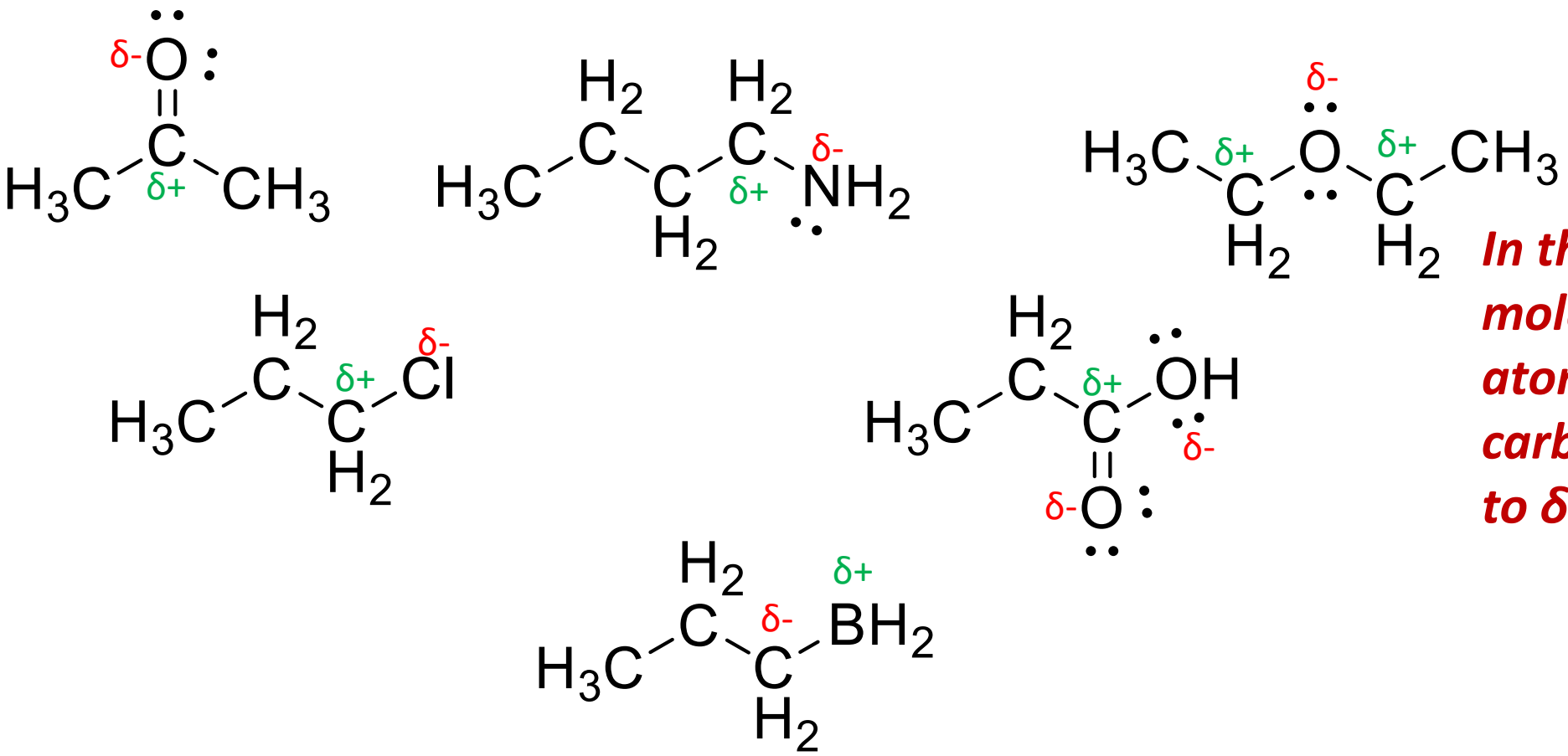
- Reaction mechanisms show how bonds break and form in terms of the flow of electrons.

REVIEW LECTURES 7 AND 8 FROM CHEMISTRY I (LEWIS STRUCTURES, POLARITY, RESONANCE, ELECTRONEGATIVITY, AND MOLECULAR GEOMETRY)!!

Electronegativity

- Recall that electronegativity is the ability of an atom to attract electrons towards itself.
- The more electronegative an atom, the more it will draw electrons towards itself.
- Electronegativity increases from left to right of the periodic table.
- The halogens are the most electronegative elements in the periodic table.
- The atom/s in a molecule (usually carbon) that are bonded to an electronegative element are said to be electropositive – electron deficient.

- More electronegative atom – represented by symbol δ-
- Electropositive atom – represented by symbol δ+

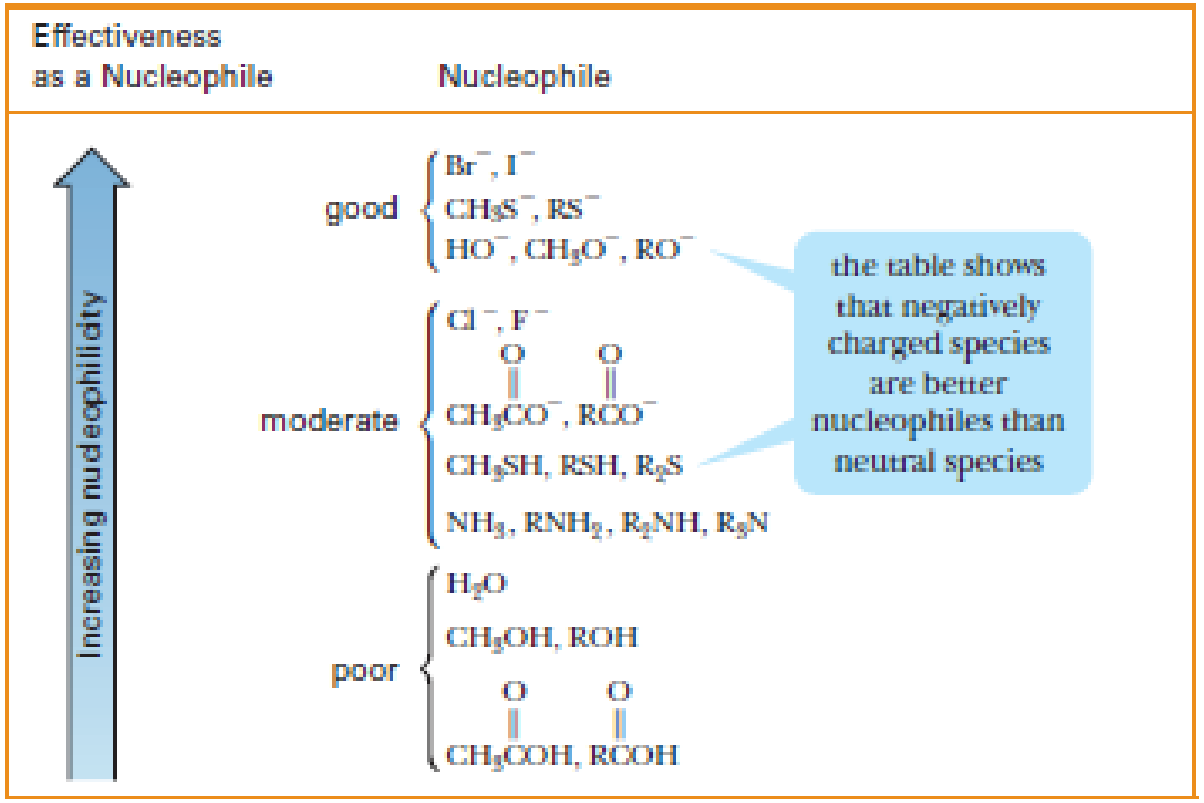


In these molecules the δ+ atoms are all carbons bonded to δ- atoms

- *Lewis acids* are species that can accept a pair of electrons to form new bonds.
- Because a Lewis acid accepts a pair of electrons, it is called an electrophile (electron lover).
- *Lewis bases* are species that have a lone pair of electrons that can be donated to form new bonds.
- A Lewis base is called a nucleophile (lover of nuclei) because it donates electrons to a nucleus with an empty, or easily vacated, orbital.

A nucleophile donates electrons. It must have a lone pair of electrons.
An electrophile accepts electrons. It must be electron-deficient.

- *Common nucleophiles*



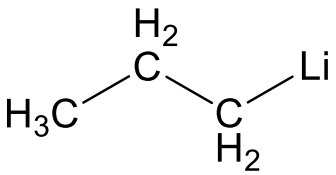
In the case of H_2O, the alcohols, and carboxylic acids, it is the oxygen which enables the molecule to act as a nucleophile due the lone pair of electrons.

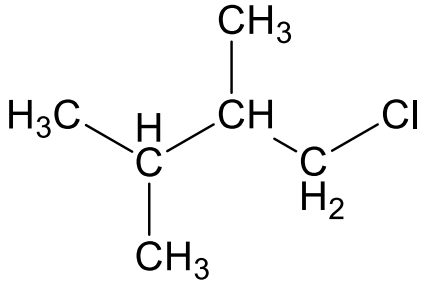
Likewise for NH_3 and other Nitrogen-containing compounds, and sulphur-containing compounds.

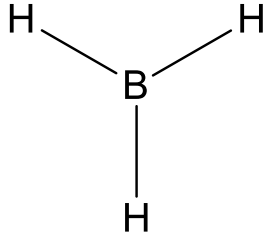
Practice exercise 3:

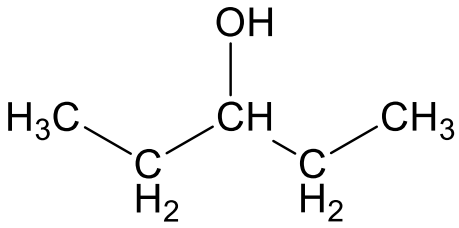
a) For the following molecules, use the appropriate symbols to indicate electronegative atoms and electropositive atoms.

b) For each molecule, identify if it can act as a nucleophile, or an electrophile.

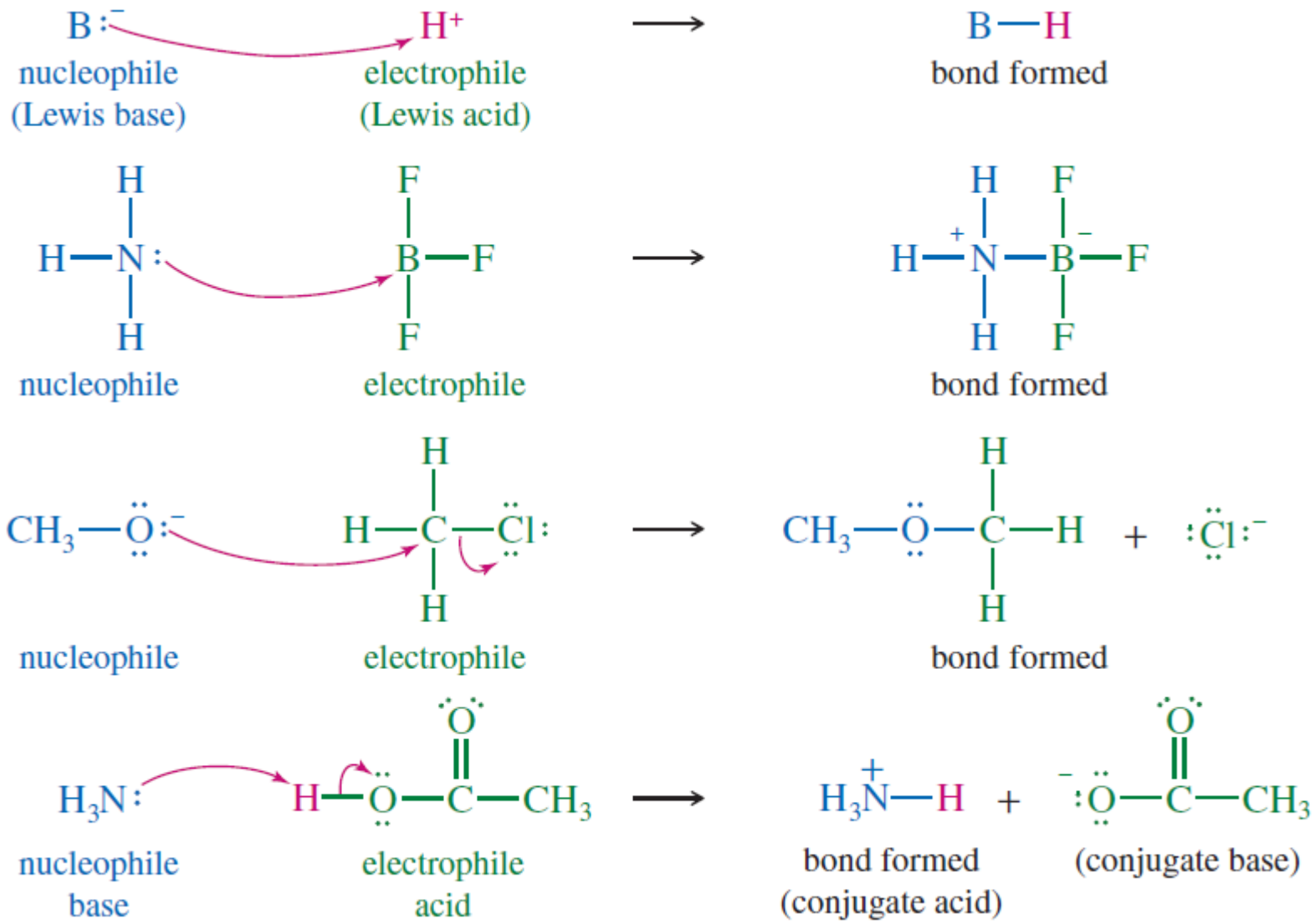








- Chemists use *curved arrows* to show the movement of electrons between nucleophiles and electrophiles.



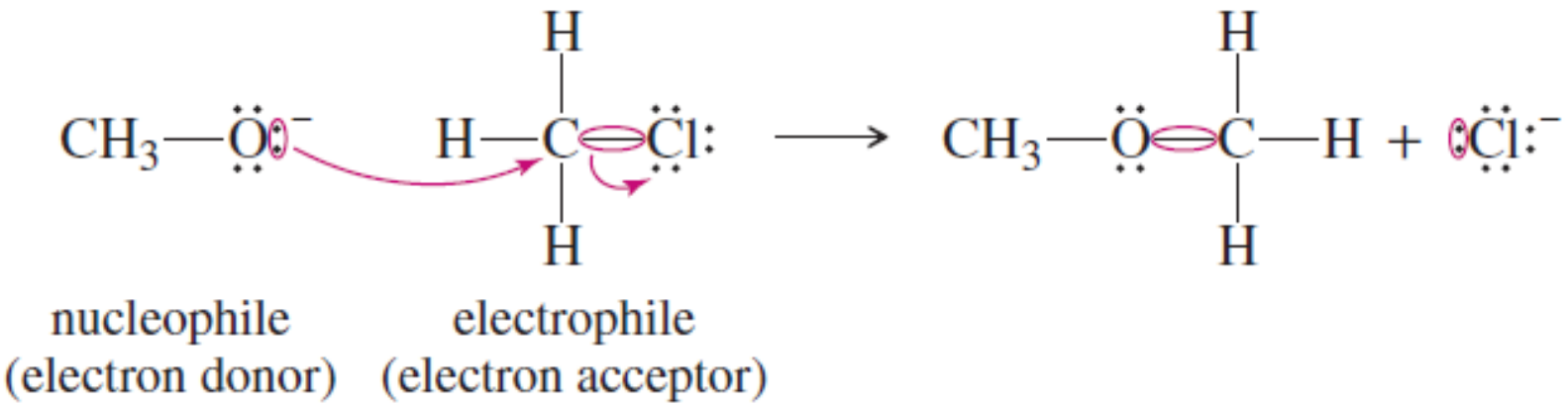
The curved arrow always starts at the nucleophile and points towards the electrophile.

- The curved arrow shows the flow of a pair of electrons from the nucleophile to the electrophile.

- It is this flow of electrons that causes bonds to form and to break.

- The movement of each pair of electrons involved in making or breaking bonds is indicated by its own separate arrow.

- For example:

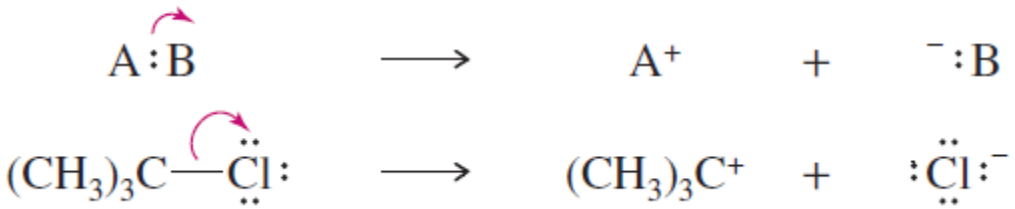


- For this reaction, one curved arrow shows the transfer of a lone pair of electrons from oxygen to carbon to form a new O—C bond. Another curved arrow shows that the C—Cl bonding pair detaches from carbon (the bond breaks), and that pair of electrons becomes a lone pair on the Cl product. This is because carbon cannot form 5 bonds!

Homolytic and Heterolytic cleavage

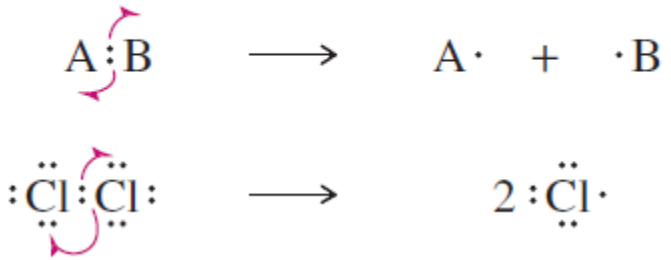
- When a bond breaks *heterolytically* (heterolytic cleavage), one atom leaves with both electrons. This results in the formation of intermediate ions.

Heterolytic cleavage (ions result)



- When a bond breaks *homolytically* (homolytic cleavage), both atoms leave with one electron each. This results in the formation of **radicals**.

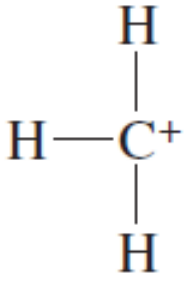
Homolytic cleavage (free radicals result)



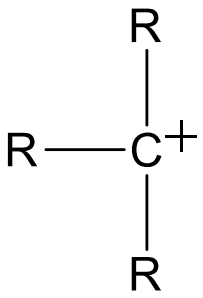
NB! Note the use of half-arrows. Half-arrows are always used to indicate homolytic cleavage (formation of radicals)

Carbocations and Carbanions

- Carbocations and carbanions are formed when bonds split heterolytically.
- A carbocation is a species containing a positively charged carbon that is bonded to three other atoms.



carbocation

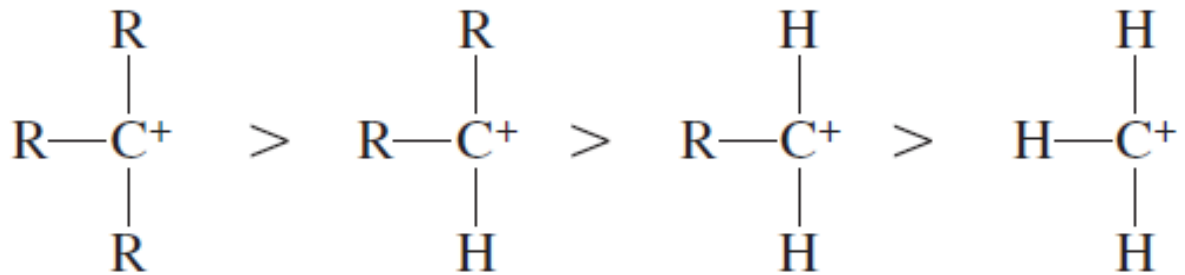


''**R**'' represents any atom / group of atoms

- They have no non-bonding pairs of electrons. They only have six valence electrons and are therefore electron deficient.
- Hence powerful electrophiles.

- Carbocations are stabilised by alkyl substituents through the **inductive effect.**
- The inductive effect is a donation of electron density from alkyl substituents through the sigma bond to the positively charged carbon.
- The more highly substituted the carbocation is, the more stable it will be.
- Generally, the more stable a carbocation is the more favourable it will be (important to remember for reaction mechanisms).

Stability of carbocations

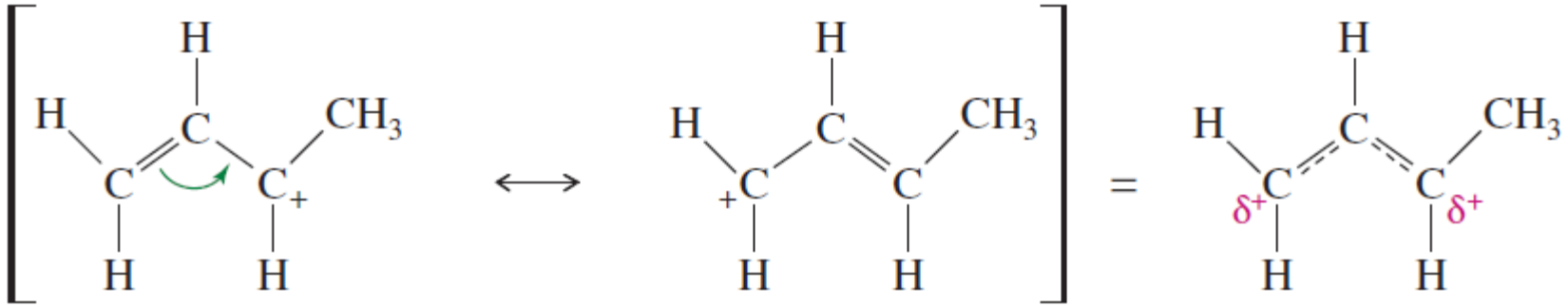


most stable — least stable

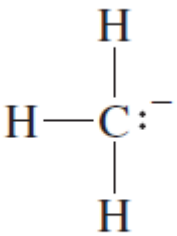
3° > 2° > 1° > methyl

'R' here represents any alkyl group

- Unsaturated carbocations are also stabilised by the inductive effect and by ***resonance stabilisation***.
- If a pi bond is adjacent to the positively charged carbon, the filled p orbitals of the pi bond can overlap with the empty p orbitals of the positive carbon.
- The result is a delocalised ion, with the positive charge being shared by two atoms. This is resonance delocalisation, and it is particularly effective at stabilising carbocations.

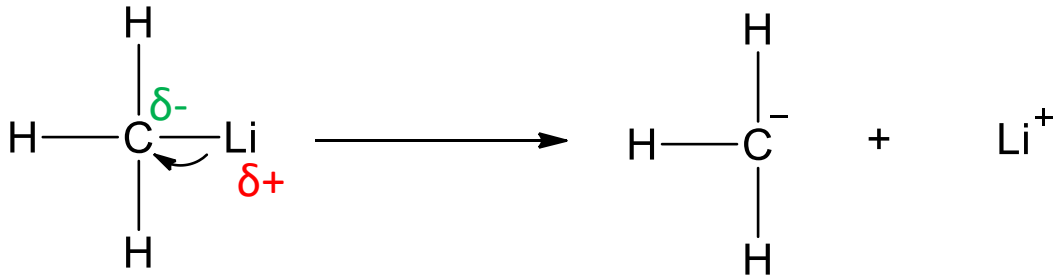


- A carbanion is a negatively charged species, containing a negatively charged carbon bonded to three other atoms.
- Carbanions form when bonds split heterolytically, and the carbon leaves with both electrons from the bond.



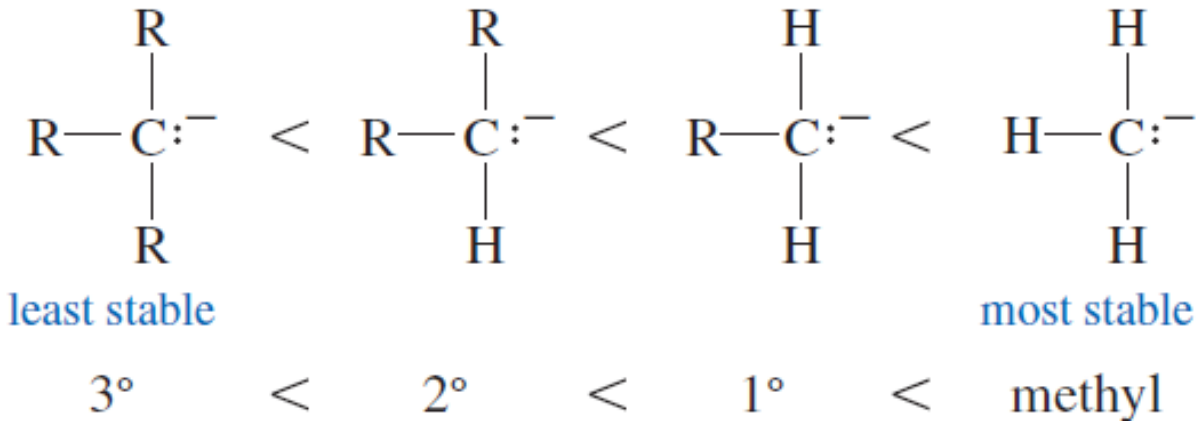
methyl anion

- They form when the carbon atom is bonded to a more electropositive atom (for example a metal).



- Carbanions are powerful nucleophiles.
- Contrary to carbocations, alkyl groups ***destabilise*** carbanions through the inductive effect.
- This is because carbanions are already rich in electron density.

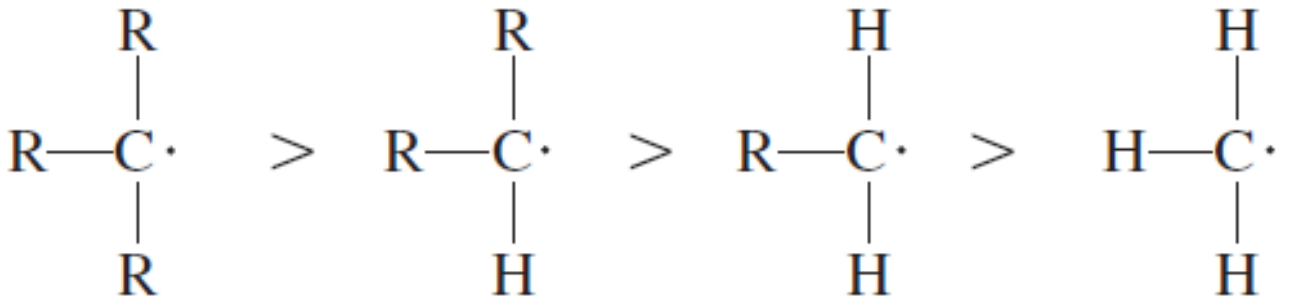
Stability of carbanions



Free radicals

- When a bond splits homolytically, free radicals are formed.
- Free radicals are neutral species that contain one non-bonding electron.
- Just like carbocations, radicals are electron deficient since they lack a complete octet around the carbon.
- Radicals are also stabilised by alkyl groups through the inductive effect.
- The more highly substituted the radical, the more stable it will be.

Stability of radicals



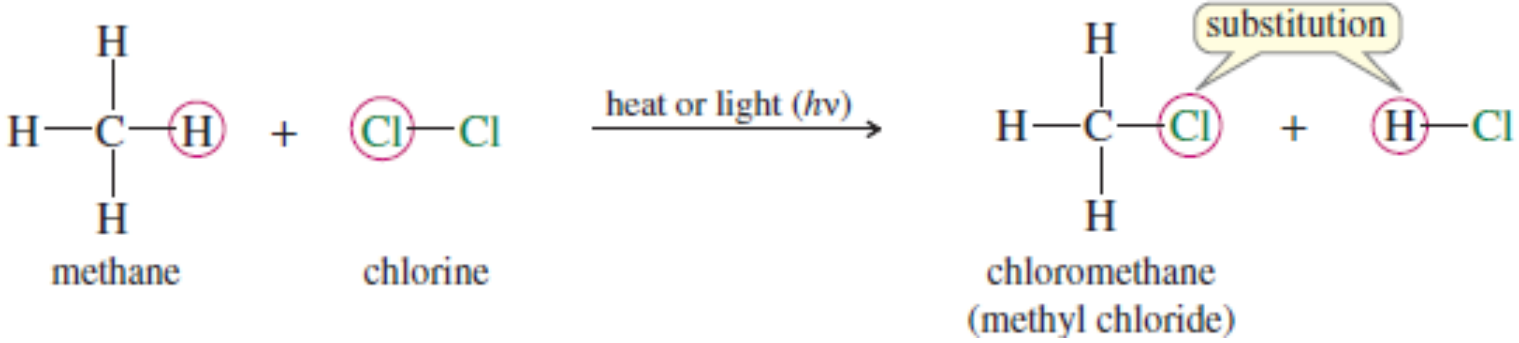
most stable — least stable

3° > 2° > 1° > methyl

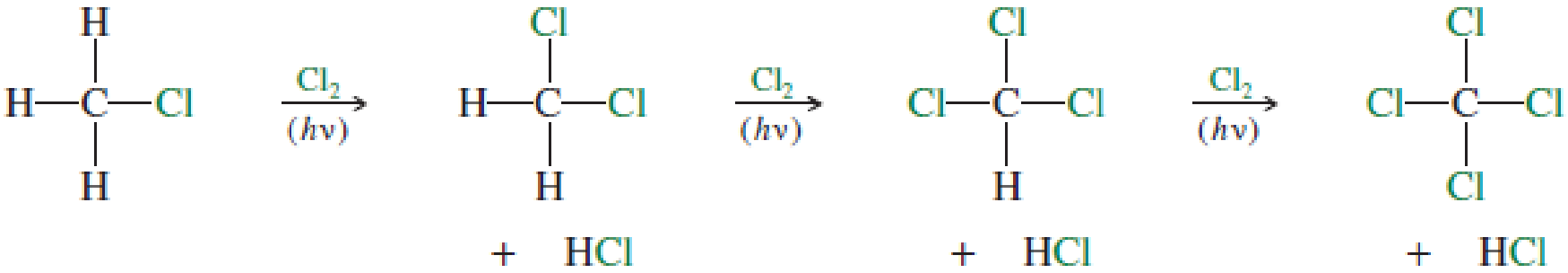
Reactivity of Alkanes

- Due to their almost complete lack of polarity, alkanes are the least reactive of the hydrocarbons.
- They will, however, undergo combustion in the presence of oxygen.
- Thus, commonly used as fuels.
- Alkanes can also undergo halogenation reactions – substitution of a hydrogen atom with a halogen.
- The reaction needs heat or light in order to take place.

- For example, chlorination of methane in the presence of heat or light;



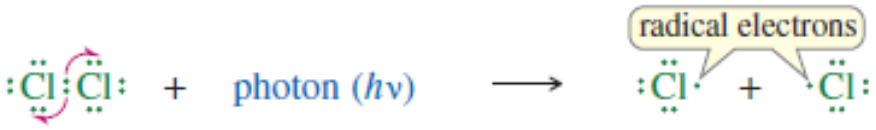
- The reaction may continue to give different chloro-substituted products; heat (ΔH) or light (*hν*) is needed for each step



- Now let’s study the mechanism of the reaction.
- The halogenation of alkanes follows a free-radical chain reaction mechanism.
- This type of mechanism has three steps;

1) Initiation step: generates a reactive intermediate
2) Propagation step: in which the reactive molecule reacts with a stable molecule to form a product, and another reactive intermediate, allowing the chain to continue until the reactant supply is exhausted or the reactive intermediate is destroyed.
3) Termination step: reactions that destroy the reactive intermediate, allowing the reaction to stop.

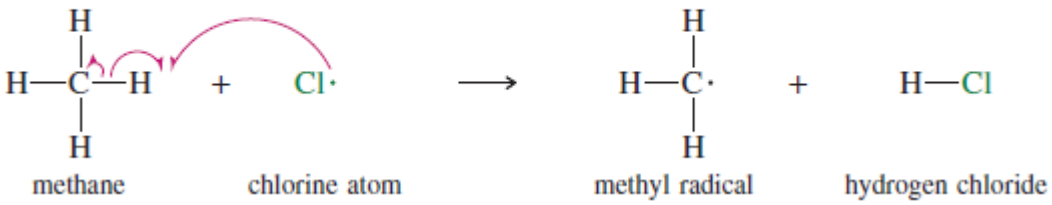
1) The initiation step: generation of radicals



Cl_2 molecule absorbs light, and splits homolytically to form two highly reactive Cl radicals. Note the use of half-arrows!

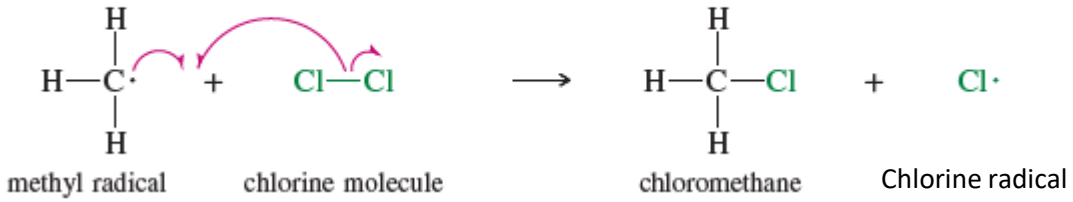
2) Propagation steps:

First propagation step



A Cl radical abstracts (removes) a proton from CH_4. the C-H bond splits homolytically, leaving one electron on the C of CH_4 forming methyl radical, and the other electron combines with Cl radical to form H—Cl.

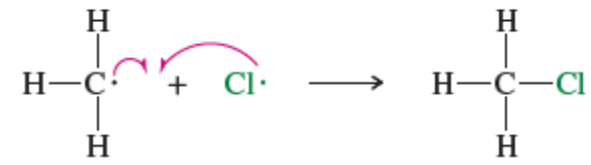
Second propagation step



The methyl radical reacts with a Cl_2 molecule, splitting the Cl—Cl bond. This generates another Cl radical that can repeat step 1.

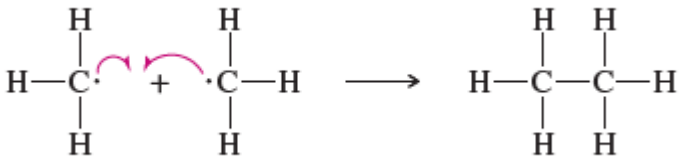
This chain of reactions continues until the supply of reactants is exhausted, or the reaction is terminated.

3) Termination step: slows or stops the chain reaction



or

or

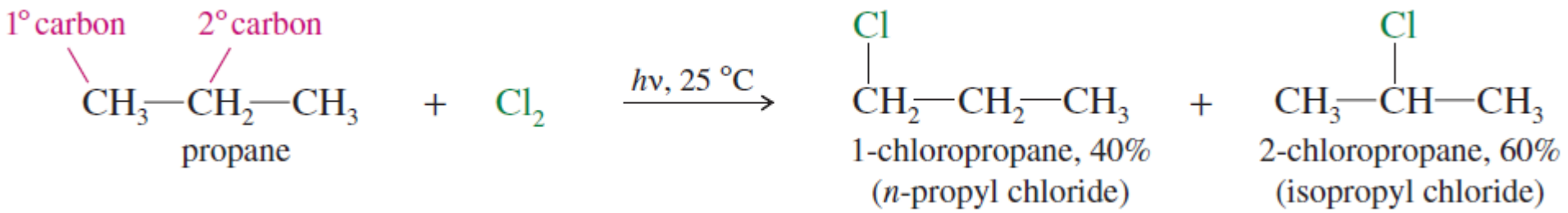


Reactive intermediate radicals are consumed; thus breaking the chain!

This termination step becomes important towards the end of the reaction, when there are relatively few molecules of reactant present.

At this point, the free radicals are less likely to encounter a reactant molecule than they are to encounter each other. The chain reaction quickly stops.

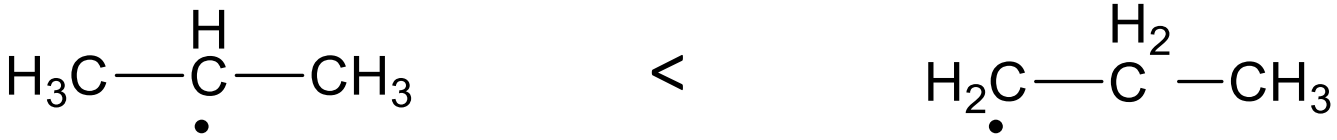
- Halogenation of alkanes containing three or more carbon atoms will result in a major and a minor product:



Minor product
40% yield

Major product
60% yield

- Due to the higher stability of a secondary radical compared to a primary radical, the secondary chloride is more favoured (major product).



More stable

Less stable